

Experiment 1

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1 Title

Study of Geiger-Muller Counter and counting statistics.

2 Aim

- I. To draw the operational characteristics curve of the Geiger-Muller counter, showing threshold voltage, operating voltage and onset of breakdown.
- II. To study the distribution curve of the counts from radioactive decay. Find mean value and standard deviation.

3 Experimental setup

3.1 Source:

We took Sodium-22 (Na^{22}) as our radiation source.

3.2 High Voltage Supply

A high voltage supply is needed for the GM counter to work. As high voltage leads to the avalanche of electrons.

3.3 Geiger Miller (GM) Counter:

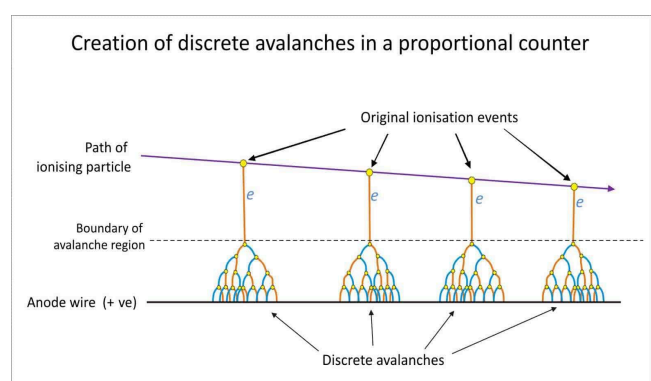
The detection of β & γ particles was carried out using a Geiger-Müller (GM) counter, which converts the ionizing radiation into measurable electrical pulses. It consists...

3.3.1 GM Tube:

The GM tube is a self-extinguishing halogen counting tube for the detection of α , β and γ radiation. It consists of a cylindrical cathode and a central anode wire filled with a low-pressure inert gas.

Townsend Avalanche:

When a radiation is passed through the GM tube, it knocks an electron from the outer orbit of that inert gas which further



hits another molecule, ionising it. This creates an avalanche under the influence of a high electric field. It's called Townsend avalanche which produces a large current pulse.

bremsstrahlung effect :

When a stray electron enters the potential field of a nucleus, it slows down and emits an UV ray, which further can start an avalanche.

3.3.2 Counting Circuit :

Each avalanche pulse is counted as a single event by the scaler-timer circuit. This gives the measure of detected beta particles per selected time range.

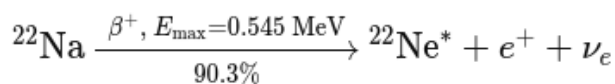
Thus whenever the HV is in good range, even one particle can start a big avalanche throughout the whole GM counter, taking it to its dead state. Which gives 1 count.

4 Theory

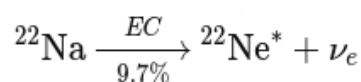
4.1 Decay Scheme of ${}_{11}^{22}\text{Na}$:

Sodium-22 shows β^+ decay where it emits β and γ rays.

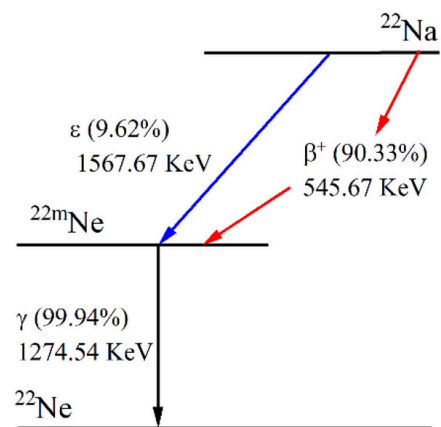
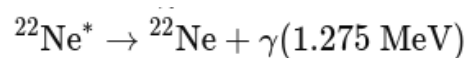
A proton in ${}^{22}\text{Na}$ converts into a neutron, emitting a positron and a neutrino, and leaves the daughter ${}^{22}\text{Ne}$ nucleus in an excited state.



The nucleus captures an inner-shell electron, turning a proton into a neutron and emitting only a neutrino, again producing excited ${}^{22}\text{Ne}$.



The excited ${}^{22}\text{Ne}$ nucleus de-excites by emitting a 1.275 MeV γ -ray, becoming stable ${}^{22}\text{Ne}$.



4.2 Gaussian Distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The normal distribution is regarded as one of the most fundamental probability distributions in statistics. This is due to several reasons. First, it is analytically tractable, meaning that a wide range of results involving this distribution can be expressed in explicit form. Second, the normal distribution naturally arises from the central limit theorem, which states that under mild conditions, the sum of a large number of random variables tends to follow a normal distribution. Finally, the characteristic “bell-shaped” curve of the normal distribution makes it an appropriate and convenient model for a wide variety of random variables encountered in practical situations.

5 Data

5.1 Part - I : Data of counts with varying HV

HV (V)	Counts			HV (V)	Counts		
	Count 1	Count 2	Count avg		Count 1	Count 2	Count avg
0	0	0	0	700	412	403	407.5
50	0	0	0	750	377	400	388.5
100	0	0	0	800	422	413	417.5
150	0	0	0	850	429	445	437
200	0	0	0	900	426	399	412.5
250	0	0	0	950	435	428	431.5
270	0	0	0	970	493	533	513
290	0	0	0	990	606	577	591.5
310	0	0	0	1010	640	658	649
330	0	0	0	1030	667	781	724
350	0	0	0	1050	838	755	796.5
400	355	370	362.5	1070	977	979	978
450	423	380	401.5	1090	1197	1191	1194
500	373	383	378	1110	1386	1536	1461
550	398	375	386.5	1120	1728	1491	1609.5
600	413	390	401.5	1130	1820	1858	1839
650	408	361	384.5	1140	2363	2165	2264

5.2 Part - II : Sorted data table of frequency of counts

We set HV as operating Voltage (675 V) found in analysis of part -1 and took 300 counts, each for 10 sec, sorted the counts and determined their frequency using python.

Count	Frequency	Count	Frequency	Count	Frequency
157	1	182	8	204	7
159	2	183	6	205	6
160	1	184	4	206	10
161	2	185	8	207	4
162	1	186	9	208	6
164	2	187	4	209	1
166	1	188	6	210	5
167	2	189	8	211	10
168	3	190	7	212	6
169	1	191	7	213	3
170	2	192	12	214	2
171	1	193	8	215	3
172	2	194	8	216	3
173	2	195	5	217	2
174	2	196	12	218	5
175	2	197	6	220	4
176	2	198	12	221	3
177	2	199	7	222	3
178	2	200	6	223	1
179	4	201	7	224	2
180	6	202	7	231	1
181	5	203	7	238	1

6 Analysis

6.1 Part - I : Operational Characteristics

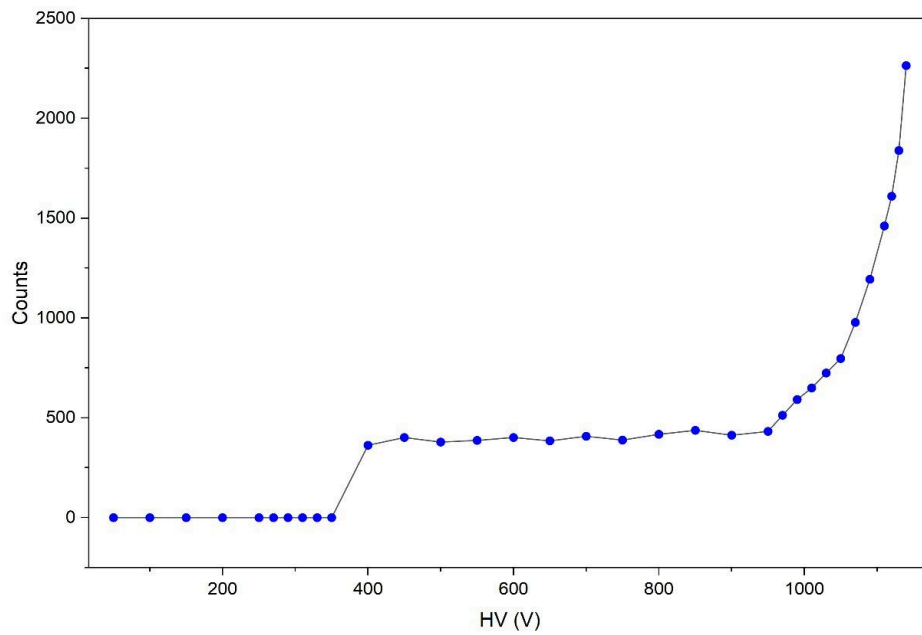


fig: Graph of the number of particle counts detected with increasing electrode voltage(V)

From this curve, we've determined :

- Threshold Voltage = **400 V**.
- Optimum / Operating HV :
It's generally defined to be in the middle of plateau region. In our data,

$$V_{op} = (400 + 950)/2 = 675 V$$

So, we analyzed the counts distribution when the GM counter is operated in **675 V**.

- Breakdown begins when continuous discharge starts: sharp, non-linear increase in counts not due to source. Here, sharp increase starts with **970 V**.

6.2 Part - II : Counting Statistics

6.2.1 Raw counts vs frequency plot : This plot is made of discrete counts, that's why it looks very **rough**.

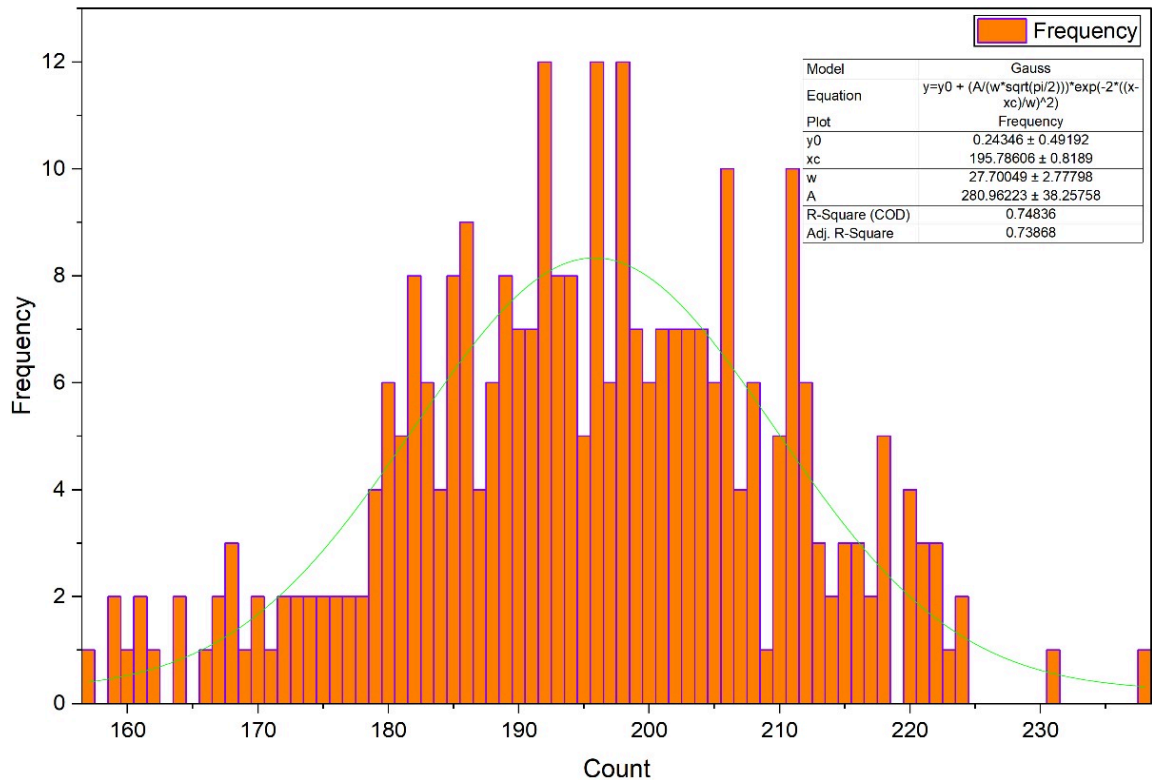


fig: Gaussian Fitted plot of Individual counts and their frequencies

From this, we get

$$\text{Median} = (196 + 196)/2 = 196$$

Minimum count = 157

Maximum count = 238

$$\text{Range of count} = 238 - 157 = 81$$

Number of counts = 300

6.2.2 Binned Frequency Table :

To get a smooth gaussian curve fit, we take counts as intervals and plot their frequencies.

Bin-width: From standard bin-width formula e.g. Sturge's rule, we get

$$k = 1 + \log_2 N = 1 + \log_2 (300) \approx 9.23 \approx 9$$

$$\text{Bin-width} = \frac{\text{Range}}{k} = \frac{238-157}{9} = 9$$

And by, summing up frequencies in each interval using python, we got this table

Counts Interval	Mean Count	Frequency
[157, 166)	161.5	9
[166, 175)	170.5	16
[175, 184)	179.5	37
[184, 193)	188.5	65

[193, 202)	197.5	71
[202, 211)	206.5	53
[211, 220)	215.5	34
[220, 229)	224.5	13
[229, 238]	233.5	2

Plotting the data,

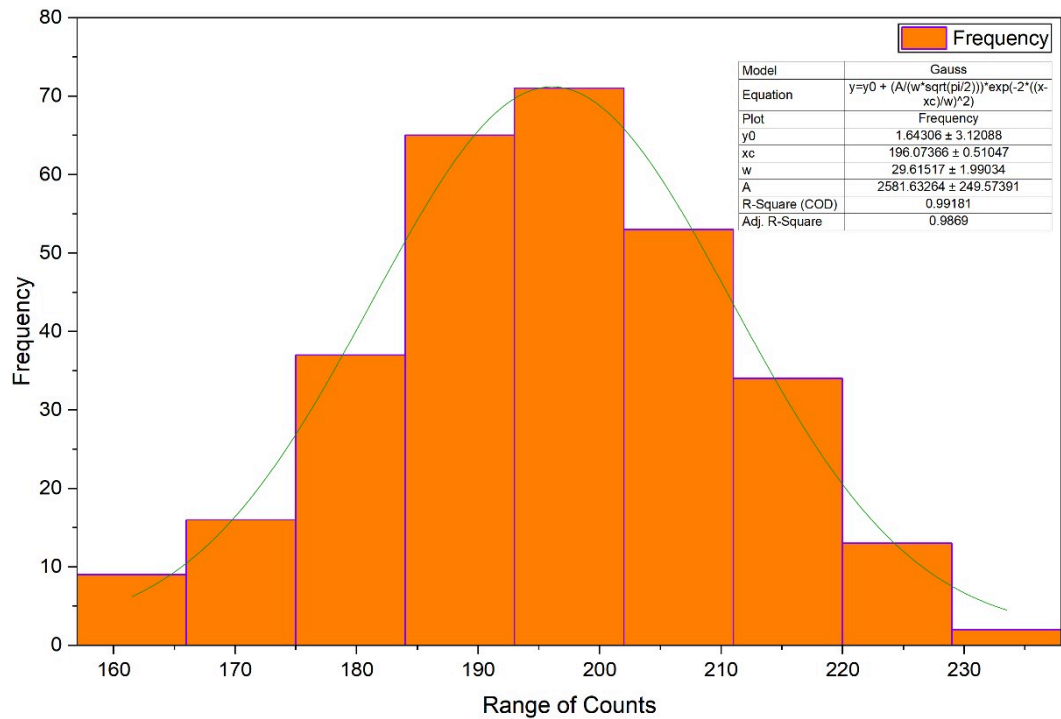


fig: Gaussian Fitted plot of Range of counts and their accumulated frequencies

6.2.3 Gaussian Curve:

After fitting, we can see the plot above indeed follows the gaussian curve. As we're not plotting discrete values, we used OriginLab's Gaussian function. 'A' here represents the area under the peak.

$$y(x) = y_0 + \left(\frac{A}{w\sqrt{\pi/2}} \right) \exp \left(-2 \left(\frac{x - x_c}{w} \right)^2 \right)$$

From our fit formula, we can get $\mu = x_c$ and $\sigma^2 = \frac{w^2}{4}$.

So, our final results are:

A	2581.632
Mean (μ)	196.073
Variance (σ^2)	219.264
Median	196
Highest Count	238
Lowest Count	157

7 Sources of Error

- Random error: In Part 1, only two runs per voltage were taken. This limited sampling leads to significant counting uncertainty, especially near the threshold and breakdown regions where counts change rapidly.
- Dead time of GM tube: At high count rates, significant counts are missed, causing underestimation of true activity.
- Background Radiation: cosmic rays, building materials, or nearby experiments and kept sources add a constant offset to all counts.

8 Conclusion

In this experiment, we successfully investigated the operational characteristics of a Geiger–Müller (GM) counter using a γ -ray source and analyzed the statistical nature of radioactive decay.